



Lesson: 1

An Introduction to Computer Networks

1. What is a Computer Network?

A computer network is a group of two or more computers or other digital devices connected together so they can share information, files, resources, and communicate with each other.

2. Example:

- Computers in a school computer lab connected together.
- Mobile phones connected through Wi-Fi.
- Computers in a bank sharing customer information.

3. What Does a Network Do?

A computer network allows users to:

- Share files and documents.
- Share printers and other devices.
- Communicate through emails, messages, or video calls.
- Access the Internet.
- Work together on the same information from different computers.

4. Why Are Computer Networks Important?

Computer networks help us:

- Save time by sharing information quickly.
- Save money by sharing resources such as





- Save money by sharing resources such as printers.
 - Improve communication.
 - Work together easily from different places.
5. Real-Life Examples of Computer Networks
- School computer lab
 - Bank network
 - Hospital network
 - Office network
 - Home Wi-Fi network
6. Points to remember:
- Network: A connection between two or more computers or devices.
 - Device: Equipment such as a computer, laptop, printer, smartphone, or tablet.
 - Internet: The world's largest computer network.

Lesson: 2

Networks cables and connectors

1. What is a Wired Network?

- A wired network is a computer network in which devices are connected using physical cables. These cables carry data from one device to another.
- A wired network is fast, reliable, and secure because the connection is made through





because the connection is made through cables instead of wireless signals.

2. Example:

- Computers connected in a school computer lab.
- Office computers connected with network cables.
- Computers connected to a router using Ethernet cables.

3. What are Network Cables?

Network cables are special cables used to connect computers and other devices in a network. They help transfer data quickly and safely.

4. What are Network Connectors?

- Network connectors are devices attached to the ends of network cables. They connect the cable to computers, switches, routers, or other network devices.

5. Advantages of a Wired Network

- Fast data transfer.
- Stable and reliable connection.

6. Disadvantages of a Wired Network

- Requires cables, making installation more difficult.
- More expensive to install in large buildings.

7. RJ-11 vs RJ-45

a. RJ-11 Connector

- Used for: Telephone networks and DSL internet connections.





- Used for: Telephone networks and DSL internet connections.

- Size: Smaller than an RJ-45 connector.

- Pins: Has 4 or 6 pins (contacts).

- Purpose: Used for voice communication and telephone line connections.

b. RJ-45 Connector

- Used for: Computer (Ethernet) networks.

- Size: Larger than an RJ-11 connector.

- Pins: Has 8 pins (contacts).

- Purpose: Used for high-speed data communication between computers, routers, switches, and other network devices.

Lesson: 3

Network safety

1. What is Network Safety?

Network safety means using computers, the internet, and computer networks in a safe and responsible way. It also includes protecting our health while using computers.

2. Health Issues When Using Computers

a. Eye Strain

- Looking at a screen for a long time can make the eyes tired or dry.

- Prevention: Take regular breaks and keep the screen at a comfortable distance.

b. Back and Neck Pain

- Sitting in the wrong position for a long time





b. Back and Neck Pain

- Sitting in the wrong position for a long time can cause pain.
- Prevention: Sit with good posture and use a comfortable chair.

c. Wrist and Hand Pain

- Continuous typing and mouse use may cause discomfort.
- Prevention: Keep wrists straight and take short breaks.

d. Headaches

- Long hours of screen time may lead to headaches.
- Prevention: Reduce screen brightness if needed and rest your eyes regularly.

3. Safety Issues When Using Networks

a. Strong Passwords

- Use strong and unique passwords.
- Never share your password with others.

b. Protect Personal Information

- Do not share personal details such as your address, phone number, or passwords online.

c. Beware of Unknown Links

- Do not click suspicious links or download files from unknown sources.

d. Use Trusted Networks

- Connect only to safe and trusted Wi-Fi networks.

e. Log Out After Use





networks.

e. Log Out After Use

- Always sign out from your accounts, especially on shared or public computers.

4. Good Computer Habits

- Sit properly while using a computer.
- Take a short break every 20-30 minutes.
- Keep food and drinks away from the computer.
- Keep your computer clean.
- Follow your teacher's or parent's instructions when using the internet.

Lesson: 4/5/6

Wireless connecting 1/2/3

Wireless Connecting 1: Wireless Network (Wi-Fi)

1. What is a Wireless Network?

- A wireless network is a network that connects devices without using physical cables.
- It uses radio waves to transmit data between devices.
- The most common wireless network is Wi-Fi.

2. Examples

- Home Wi-Fi
- School Wi-Fi





- School Wi-Fi
- Office Wi-Fi
- Public Wi-Fi in airports and cafés

3. Advantages

- No cables are required.
- Easy to install and use.

4. Disadvantages

- Can be affected by walls or interference.
- Less secure if not password protected.

Wireless Connecting 2: Connecting a Device Using Bluetooth Technology

1. What is Bluetooth?

- Bluetooth is a wireless technology used to connect nearby devices over a short distance without using cables.

2. Common Uses

- Connecting wireless headphones or speakers.
- Sharing files between phones.
- Connecting a wireless keyboard or mouse.
- Connecting a smartwatch to a smartphone.

3. Steps to Connect Using Bluetooth

Step 1. Turn on Bluetooth on both devices.

Step 2. Make the device discoverable.

Step 3. Search for available devices.

Step 4. Select the device you want to





Step 4. Select the device you want to connect.

Step 5. Confirm the pairing request.

Step 6. The devices are now connected.

4. Advantages

- No cables needed.
- Easy to use.
- Low power consumption.

5. Disadvantage

- Short range (about 10 meters).
- Slower than Wi-Fi for large file transfers.

Wireless Connecting 3:

Mobile Hotspot

1. What is a Mobile Hotspot?

- A mobile hotspot is a feature that allows a smartphone or tablet to share its mobile internet connection with other devices such as laptops, tablets, or other phones using Wi-Fi.

2. How to Use a Mobile Hotspot

- Turn on Mobile Data.
- Open Settings.
- Select Mobile Hotspot or Personal Hotspot.
- Turn on the hotspot.
- Set a Wi-Fi name and password.
- Connect another device using the hotspot name and password.

3. Advantages





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- Turn on the hotspot.
- Set a Wi-Fi name and password.
- Connect another device using the hotspot name and password.

3. Advantages

- Internet can be shared anywhere with mobile network coverage.
- No router is needed.
- Easy to set up.

4. Disadvantages

- Uses mobile data.
- Drains the phone's battery quickly.

In short:

- A Wireless Network (Wi-Fi) connects devices without cables.
- Bluetooth connects nearby devices for file sharing and accessories.
- A Mobile Hotspot shares a smartphone's internet connection with other devices.

